Door Lock Control System - High-Level Design

Document version 1.1

Status: released for review

1 Introduction

This High-Level Design (HLD) describes the software architecture for locking and unlocking of the four vehicle doors from the key fob, the interior switch and the vehicle speed. It lists the software components, their ports and interfaces, the signals exchanged between them and the main functional flows.

The document is the reference for architects, developers, integrators and test engineers. Names of components, interfaces and signals are written exactly as they appear in the software architecture model.

1.1 Scope and assumptions

The design covers the application layer only. Basic software, bootloader and network management are described in separate documents. All components run in the same ECU and communicate through the runtime environment (RTE) using sender-receiver ports.

It is assumed that every signal has exactly one producer. Each signal is refreshed in the cycle time given in the signal list.

1.2 Terms

Term	Meaning
SWC	Software component, the smallest deployable unit of application software.
Port	Connection point of a component. Provided ports send data, required ports receive data.
Interface	Typed description of the data elements exchanged over a port.
Signal	Named data item with one producer and one or more consumers.
RTE	Runtime environment that connects the components.

2 Architecture Overview

The system consists of 5 software components. Each one has a clearly defined responsibility, and the dependencies between components are kept as small as possible. The following sections give the details of every component.

Components that provide data are placed close to the hardware abstraction, while controller components sit above them and make the decisions.

3 Software Components

3.1 Swc_DoorLockCtrl

Decides whether the doors must be locked or unlocked. It combines requests from the key fob receiver and the interior switch with the current vehicle speed, and enables speed-dependent auto lock above 10 km/h.

Swc_DoorLockCtrl has 3 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Rp_KeyFobReq	required	If_KeyFobRequest
Rp_VehSpeed	required	If_VehSpeed
Pp_LockCmd	provided	If_DoorLockCmdV2

Dependencies: Swc_VehSpeedProv, Swc_KeyFobRx.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.2 Swc_KeyFobRx

Receives the rolling-code message from the key fob, checks the counter window and forwards a validated request.

Swc_KeyFobRx has 1 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Pp_KeyFobReq	provided	If_KeyFobRequest

Dependencies: none.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.3 Swc_VehSpeedProv

Provides the filtered vehicle speed that is calculated from the wheel speed signals of the brake ECU.

Swc_VehSpeedProv has 1 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Pp_VehSpeed	provided	If_VehSpeed

Dependencies: none.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.4 Swc_LockActuatorDrv

Drives the lock motor of each door, monitors the end stops and reports the actuator state back to the controller.

Swc_LockActuatorDrv has 3 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Rp_LockCmd	required	If_DoorLockCmdV2
Pp_ActuatorStatus	provided	If_ActuatorStatus
Rp_AjarState	required	If_DoorAjar

Dependencies: Swc_DoorAjarMon.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.5 Swc_DoorAjarMon

Monitors the door contacts and reports whether each door is closed.

Swc_DoorAjarMon has 1 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Pp_AjarState	provided	If_DoorAjar

Dependencies: none.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

4 Interfaces

All interfaces are sender-receiver interfaces. The data elements of each interface are listed below.

Interface	Type	Data elements
If_KeyFobRequest	Sender-Receiver	LockRequest, UnlockRequest, PanicRequest
If_VehSpeed	Sender-Receiver	VehSpeedKph, VehSpeedValid
If_ActuatorStatus	Sender-Receiver	MotorState, EndStopReached
If_DoorAjar	Sender-Receiver	DoorFL, DoorFR, DoorRL, DoorRR
If_DoorLockCmdV2	Sender-Receiver	LockTarget, UnlockTarget, Timeout, RetryCount

5 Signal List

Producer and consumers of every signal are given in the table. A dash means that no consumer is defined.

Signal	Producer	Consumers	Cycle
Sig_DoorLockState	Swc_LockActuatorDrv	Swc_DoorLockCtrl	20 ms
Sig_KeyFobValid	Swc_KeyFobRx	Swc_DoorLockCtrl	20 ms
Sig_VehSpeedKph	Swc_VehSpeedProv	Swc_DoorLockCtrl	20 ms
Sig_DoorAjar	Swc_DoorAjarMon	Swc_LockActuatorDrv	20 ms

6 Functional Flows

6.1 Unlock by key fob

Step 1: The key fob sends an unlock request.

Step 2: Swc_KeyFobRx validates the rolling code and sets Sig_KeyFobValid.

Step 3: Swc_DoorLockCtrl checks the vehicle speed and sends Pp_LockCmd.

Step 4: Swc_LockActuatorDrv moves the motor and reports Sig_DoorLockState.

6.2 Speed dependent auto lock

Step 1: Swc_VehSpeedProv publishes Sig_VehSpeedKph every 20 ms.

Step 2: When the speed stays above the threshold for 2 s, Swc_DoorLockCtrl requests a lock.

7 Revision History

Version	Date	Change
1.0	2026-03-02	Initial release
1.1	2026-05-20	Added door ajar monitor, renamed lock command interface, removed lock fault flag